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Evaluating De-Escalation Training for Direct and Indirect Employees Caring for Residents With Huntington's Disease

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Brandon Rogers[1], Barbara St. Marie[2], Daniel Wesemann[3], and Peg Nopoulos[4]

Abstract

INTRODUCTION: Little is known about reducing the challenges for caregivers and patients with Huntington's disease (HD). HD creates behavioral disturbances, cognitive decline, and motor disorder progression over the lifetime requiring some individuals to need long-term facility care. **AIMS:** There are concerns about safety and confidence of employees caring for residents with HD. **METHODS:** Nursing staff, administrators, and auxiliary employees were recruited from a long-term care (LTC) facility in rural Iowa, from July 2020 to August 2020. A de-escalation training intervention was delivered. The 1-day intervention included resident behaviors, planning and safety, teamwork, communication, and included role play and simulation. A pre- and post-survey measured confidence and competence in caring for people with HD before and after a training intervention. A resident medical record audit explored challenging behaviors before and after the training intervention. **RESULTS:** Of 25 participants, six were registered nurses/licensed practical nurses (RNs/LPNs; 24%), four administrators (16%), eight nursing assistants (32%), and seven auxiliary employees (28%). There was improvement in employees perceived safety (33.3%), co-workers enjoyment working with HD residents (54%), understanding symptoms of HD (44.4%), confidence in job abilities (21.0%), and confidence in ability to care for patients with HD (26.3%). A medical record audit showed decreased documentation of resident aggression and care refusal post-intervention. **CONCLUSIONS:** These findings suggest de-escalation training in LTC facilities increased perception of job safety, co-workers' enjoyment, understanding HD symptoms, confidence in ability to care for patients with HD, and decreased resident agitation and care refusal.

Keywords

aggression, behavior/cognitive behavior therapy, community residences, quality improvement, safety

Introduction

Problem Description

Huntington's disease (HD) is a long-term fatal, autosomal dominant genetic neurodegenerative disorder that causes motor, cognitive, and psychiatric symptoms. In the United States, there are approximately 41,000 individuals who are symptomatic with an additional 200,000 who have a 50/50 chance of inheriting HD (Huntington's Disease Society of America [HDSA], 2021b). The typical age of onset is between 30 and 50 years; and individuals with HD have a prognosis of 10 to 20 years that might eventually require 24-hour care (HDSA, 2021b).

Symptoms of HD are caused by a trinucleotide repeat expansion on chromosome four called the "Huntington Gene" (HTT Gene; McClogan & Tabrizi, 2017). The non-pathological HTT gene repeats between 17 and 35 times, and pathology occurs when the HTT gene repeats beyond 40 times, causing symptoms later in life. Atrophy of the striatal neurons occurs first, then degeneration of the entire neuronal network of the brain leads to involuntary choreiform movements, cognitive impairment, and

[1]Brandon Rogers, DNP, ARNP, PMHNP-BC, University of Iowa Hospitals & Clinics, Iowa City, IA, USA
[2]Barbara St. Marie, PhD, AGPCNP, FAANP, FAAN, University of Iowa, Iowa City, IA, USA
[3]Daniel Wesemann, DNP, MSW, PMHNP-BC, FAANP, University of Iowa, Iowa City, IA, USA
[4]Peg Nopoulos, MD, University of Iowa Hospitals & Clinics, Iowa City, IA, USA

Corresponding Author:
Barbara St. Marie, Associate Professor, College of Nursing, University of Iowa, 50 Newton Road #440, Iowa City, IA 52242, USA.
Email: Barbara-stmarie@uiowa.edu

psychiatric symptoms (HDSA, 2021b; McClogan & Tabrizi, 2017). Falls, high risk for aspiration, choking, loss of ability to complete activities of daily living (ADLs), refusal of care, and verbal and physical threats/assaults create safety risks for patients and care givers. Admittance to long-term care (LTC) facilities is often needed where a variety of medical and non-medical services can be provided. However, these facilities are already impacted by higher levels of occupational injuries compared with other types of facilities (U.S. Bureau of Labor Statistics, 2015). In 2014, there were 7.1 recordable occupational illnesses and injuries per 100 full-time equivalents in LTC (Lapane et al., 2016).

Available Knowledge

Finding capable LTC facilities is challenging for individuals with late-stage HD and their families. Studies show that a range of 2.0% to 7.6% of patients with HD is cared for in LTC (HDSA, 2021a). Yet, there is a lack of specialty units for those with HD (McClogan & Tabrizi, 2017). The United States has approximately 20 LTC facilities with HD specialty units (HDReach, 2022). In 2018, a rural LTC facility in central Iowa affiliated with the University of Iowa Hospitals and Clinics transitioned from general LTC to a HD specialty care facility and the facility required training to provide quality care for individuals with HD (McClogan & Tabrizi, 2017).

Rationale

De-escalation training helps employees learn crisis prevention and management skills through behavioral competencies (Gaynes et al., 2016). The de-escalation training curriculum was based on an Applied Behavioral Analysis (ABA) framework, using positive and negative reinforcement, behavioral modification, and verbal de-escalation techniques to enable employees to apply the best prevention interventions (Cooper et al., 2020; Couvillon et al., 2010). ABA strategies have been successful in improving cognitive performance, language skills, and adaptive behavior skills for populations, such as individuals diagnosed with autism spectrum disorders (Warren et al., 2011), attention-deficit hyperactivity disorder, rehabilitation for people with brain injury, substance use disorder, and intellectual disabilities (Couvillon et al., 2010). The de-escalation program implemented was found to be easily accessible and supported by the University of Iowa Hospitals and Clinics, a Huntington's Disease Society of America's Center of Excellence, and the de-escalation techniques were not previously used by the facility in this study.

Specific Aims

The purpose of this project was to implement a de-escalation training for both direct and indirect care employees at a rural central Iowa LTC facility to improve employees' knowledge and confidence in their ability to care for residents with HD, reduce residents' harmful behavioral episodes, and improve employees' perception of safety on the job.

Methods

Context

This quality improvement study recruited employees from a recently designated HD LTC facility in rural Iowa. Employees included registered nurses (RNs), certified nursing assistants (CNAs), nursing administration and auxiliary employees (e.g., maintenance, housekeeping, recreation therapy, kitchen/dietary, laundry). Inclusion criteria were (1) 18 years of age or older, (2) employed by the LTC facility, (3) direct or indirect contact with residents with HD, (4) English-speaking, and (5) willing to participate in a 1-day, 8-hour course. Participation was voluntary and participants received usual hourly pay by the facility with no requirement of over-time.

Intervention

De-escalation training intervention occurred for employees at an LTC facility. This 1-day, 8-hour course contained content for resident behaviors, planning and safety, teamwork, communication, and included role-play and simulation. During the day, there were two 10-minute breaks, a 30-minute lunch, and was taught in July 2020 by one instructor (BR). Four courses were offered with six to seven participants in each course. Consistent with facility goals and individual needs, the de-escalation program was written, so that, there were no hands-on escorts nor restraints, rather it did ensure specific focus on physical considerations and symptoms related to HD, latest research, and shared experiences of a family at risk for HD. Each participant received an 85-page manual, teaching methods were didactic with lecture, group discussion and simulation with verbal de-escalation. COVID-19 restrictions were followed at the facility per CDC, 2020 (Centers for Disease Control and Prevention, 2020) guidelines.

The Administrator of the LTC facility approved this training and compensation for employees in attendance. A mentor team (DW, BS, and PN) was available to the lead author for study design, discussions of implementing during a pandemic, and preparation of this article.

Study of Interventions and Measures

The measurement approach chosen for assessing the impact of the intervention were surveys and chart audit. The pre-survey staff measurements and the chart audit occurred 30 days prior to the training intervention and 30

days post-intervention to determine whether outcomes were associated with the intervention.

Surveys. Prior to intervention, the participants completed a nine-item, 5-point Likert-type scale survey ranging from "strongly disagree" to "strongly agree" to assess three domains: satisfaction, knowledge, and confidence. The nine items within the three domains were about perception of LTC employees' safety, participants' perception of their co-worker's enjoyment while caring for residents with HD, participant understanding of HD symptoms, participant confidence in job abilities and confidence in care for residents with HD, participant's confidence in co-workers while observing their care for residents with HD, participant job satisfaction, and whether they would want more education about HD and de-escalation of behavioral episodes. (Survey is available upon request.) The survey was developed by the first author (BR) who has 24 years of experience in caring for people with HD (personal and professional). Face validity was obtained by content expert (PN) review. In addition, the project committee members (DW, BS, and SM) reviewed and approved the survey items for this study. Post-intervention, participants received a self-addressed stamped envelope containing the same survey and instructions to mail the completed survey to the course instructor (BR).

Chart Audit. A chart audit was used to find documentation that reflected potentially harmful resident behaviors and the frequency of these documented words. Words (selected by the first author) for use in the chart audit included: aggression, aggressive, agitated, angry, acting out, disruptive, refused, threatening, assault, yelling, physical or verbal assault, as needed medications, refusal of meals, refusal of medications, refusal of ADLs, and refusal of ambulation.

Analysis

Descriptive analysis of the surveys included comparison of the percentage of participants who reported feeling satisfied, knowledgeable, and confident in their ability to care for residents with HD before and after the intervention. The chart audit was analyzed based on a comparison of the frequency of designated words found in the period before and after the intervention. Words were then grouped into three categories: aggression, agitation, and refusal. "Aggression" included: threatening, aggression, aggressive, assault, and physical or verbal assault. "Agitation" included: angry, acting out, agitation, disruptive, or yelling. "Refusal" included: refused/refusal of meals, medications, ADLs, or ambulation. Medication administration was also documented in residents' charts to reduce unsafe or challenging behaviors, so that, the frequency of "PRN" or "as needed medications" documentation in the medical record was counted.

Ethical Considerations

This study was deemed not human subjects research by the Institutional Review Board at the University of Iowa.

Results

One hundred percent of the employees ($N = 25$) volunteered to participate in de-escalation training on-site and completed pre- and post-surveys in July and August 2020. There was a 100% return on 30-day surveys. Participants included six RN/licensed practical nurses (LPNs), four administrators, eight nursing assistants, and seven auxiliary staff. Years of experience ranged from 3.5 to 5.7 years. At the time of the study, the facility was home to 13 residents with HD (50% of total residents). One of the residents expired during the study and data from their chart audit was included.

Intervention Results

Pre-Intervention Survey. Figure 1 depicts employees' survey responses prior to the intervention and shows areas where improvement was needed. These areas were: feeling safe doing their job ($n = 12$), understanding symptoms of HD ($n = 10$), and their co-workers enjoyed caring for HD residents ($n = 6$). Employees indicated they needed more education on de-escalation training ($n = 17$), and on behavioral management of the residents with HD ($n = 19$).

Post-Intervention Survey (30 Days Post-Intervention). Results showed no change in job satisfaction, but self-reported safety, knowledge, and self- and co-worker confidence increased from before to after the intervention (see Figure 1, pre- and post-intervention survey results). Specifically, there was improvement in feeling safe doing their job (33.3% increase), understanding HD symptoms (44.4% increase), confidence in general abilities caring for residents at the facility (21.0% increase), and specifically caring for residents with HD (26.3% increase). Participants were asked to observe their co-workers caring for those with HD, and reported a 5% improvement in confidence in co-workers' ability to care for residents with HD, and a 54% increase in perception of co-worker's enjoyment in caring for residents with HD. Post-intervention, there was 5% reduction in requests for more education on de-escalation management and a 16.0% reduction in more education needed for working with behaviors associated with HD.

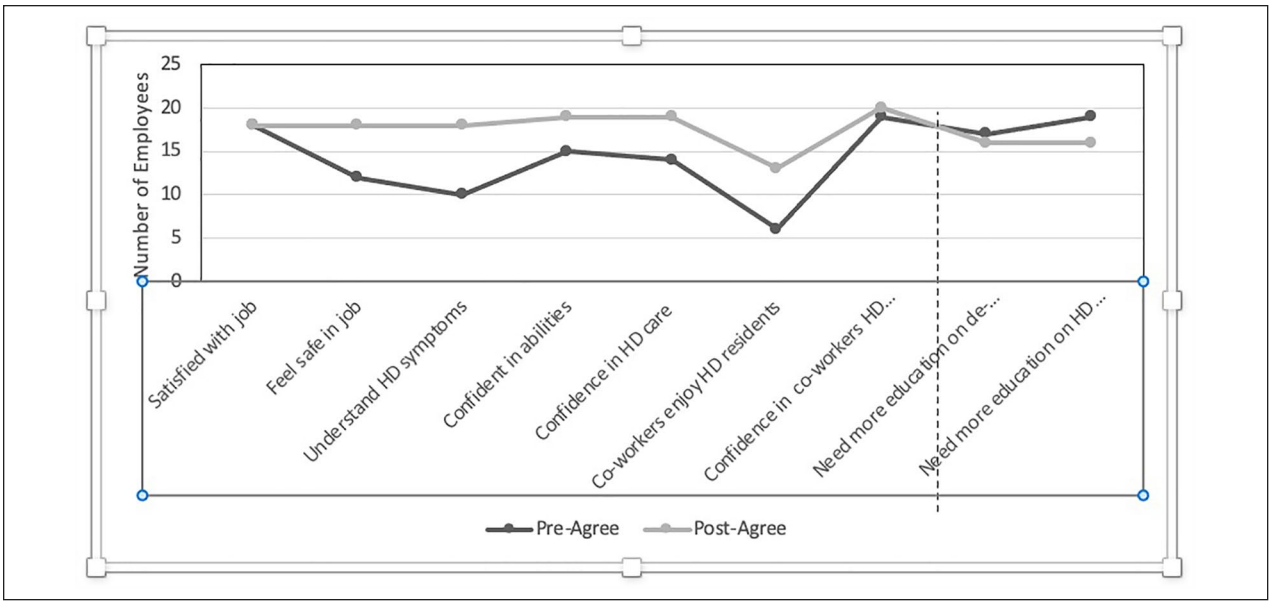


Figure 1. Pre- and Post-Intervention Survey Results.
Note. Last two measurements for pre- and post-pertain to questions answered by participants. HD = Huntington's disease.

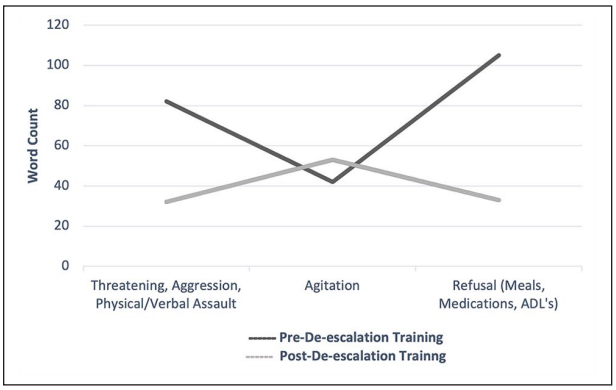


Figure 2. Chart Audit for Behavior Documentation.
Note. ADL = activity of daily living.

Post-Intervention Chart Audits. A comparison of pre–post documentation showed improvement in documentation regarding resident behavior (see Figure 2, chart audit for behavior documentation). There was a 61.0% reduction in notations of aggression by the resident toward the employees, and a 68.6% reduction in documentation for refusal of care by the residents following the intervention. Notations reporting agitation increased 26.2%, and documentation reporting "as needed" medications administered for unsafe or challenging behaviors increased 48%.

Discussion

Through implementation of de-escalation training there was no change in job satisfaction, however, we identified an increase of 33.3% in perception of LTC employees' feeling safe in their jobs, an increase of 54% of perceived co-worker enjoyment caring for HD residents, an increase of 44.4% of participants' understanding HD symptoms, and an increase of 21% confidence in job abilities and care for residents with HD. We also found a decrease in documented residents' aggressive behavior and refusal of care, such as meals, medication, ADLs, and ambulation. Employees receiving the training were multidisciplinary, including RNs, LPNs, CNAs, administration, housekeeping, recreational therapy, kitchen/dietary, and laundry personnel. Several of our findings are important for further discussion.

Employees Safety

Unsafe challenging behaviors can create safety concerns for employees caring for residents with HD. Legislation to support safe resident handling in nursing facilities has been enacted in several states in the United States since 2005 (Lapane et al., 2021; National Institute of Nursing Research/National Institutes of Health [NINR], 2022; U.S. Department of Labor, 2022), yet worker injuries continue to occur. Prior to training, caregivers expressed concerns over safety while caring for residents with HD when agitation or aggression occurred. Prevalence of agitation in individuals with HD range from 22% to 66% (McCloskey et al., 2017) and at times acute agitation rapidly escalates (Craufurd et al., 2001).

After completion of the de-escalation training, our study showed a 61% reduction in documentation of

aggression by residents toward employees, however, our documentation showed a 26.2% increase in reporting agitation. Other studies have also indicated increases in reporting violence/agitation in health care following de-escalation training and explained these increases as greater participant awareness rather than actual increases in patient aggression (Arnetz & Arnetz, 2000; Celofiga et al., 2022; Deudon et al., 2009; Leach et al., 2019; Livingston et al., 2009).

Psychotropic medications may contribute to employee and patient safety. The use of psychotropic medications is not well studied and lacks standardization but remains a prominent component of care when a potential medical or non-medical condition for agitation has been ruled out (Craufurd et al., 2001). The 1-hour didactic training during our intervention provided information for managing symptoms of residents with HD and use of selected PRN medications while limiting polypharmacy. There was a 48% increase in PRN medication administration following de-escalation training. While this program is behaviorally focused and teaches recognition of escalating behavior, the didactic portion of the curriculum provided additional content on proactive use of medications to help limit agitation and aggressive episodes (HDSA, 2021a) and may account for this increase in PRN medication administration.

Co-Worker Enjoyment Caring for HD Residents

An important finding was the employees' perception that their co-workers enjoyed caring for residents with HD (increase of 54%) following training. Research shows that trust and mutual respect among co-workers is important to strengthen effective communication and contribute to reaching common goals in caring for patients (Rossi & Oh, 2020). This trust and positive work environment reduces stress, improves employee's compliance with procedures, and leads to greater safety for patients and employees (Vassbø et al., 2019). This valued recognition of co-workers likely contributes to the effectiveness of this training in improving clinical practice.

Understanding Management of Symptoms of Residents With HD

There is a lack of understanding and research on management of symptoms of HD in LTC facilities (Craufurd et al., 2001; Rossi & Oh, 2020). The addition of a family member's perspective to the de-escalation training helped to promote employee discussions during this portion of the training. Despite the many challenges caring for those in later stages of HD, employees felt that they understood symptoms of HD and were more confident in their ability to de-escalate when challenging behaviors occurred.

Confidence in Their Job and Ability to Care for Residents With HD

In 2018, the study facility transitioned from LTC to a facility that specializes in HD and LTC. Employees reported resistance to the transition, however, after this training they felt more confident in caring for residents with HD. A web-based national survey of patients with HD and caregivers ($N = 797$) showed one in five people with HD were currently not receiving community nor medical care (Anderson et al., 2019). This finding indicated a significant gap in providing safe and effective care to patients with HD and their caregivers. Our study showed a 33.3% increase in employees feeling safe on the job, indicating improvement in self-reported safety. This was consistent with another study using a standardized and evidence-based educational intervention focused on patients and caregivers, resulting in improvement of HD symptoms and anxiety by patients and less psychosocial burden of the caregivers (A'Campo et al., 2012). These studies support requisite education of caregivers in LTC facilities to improve employees' confidence in their ability to care for residents with HD, reduce unsafe behavioral episodes, and improve employees' perception of job safety.

Changes for the Future

This educational intervention aligns with the strategic mission of NINR. The mechanisms within the NINR are to promote the health of "individuals, families and communities" (NINR, 2022). Also, supportive of this intervention is the National Institute of Occupational Safety and Health Administration (OSHA). Their mission is "to ensure safe and healthful working conditions for workers" (U.S. Department of Labor, 2022). Standardized and evidence-based education supports the mission of these highly regarded institutions. It is imperative that access to this type of education for LTC employees can ensure new knowledge and confidence in providing care for resident with a complex, multifactorial disease, such as HD. This project is unique, in that, education was provided not only to all levels of nurse employees, but also to ancillary employees who have frequent interactions with residents but limited clinical training. Implementation of de-escalation training that is supported by research with favorable outcomes could help improve employees' job satisfaction, knowledge, and confidence in their care of residents with HD and confidence in their co-workers' care of residents with HD.

Sustainability of up-to-date training at the facility will require an investment of employees and administration (U.S. Department of Labor, 2022). Organizations must dedicate financial resources to sustain a de-escalation trainer on-site and to ensure continued educational support for updated treatments and research. Affiliation with an area hospital can further enhance this investment of annual training and refresher courses by spreading the cost of hiring an expert or training-the-trainer among systems of care.

Limitations

This study had several limitations. Data collection did not include the types of PRN medications or dosage. This limitation is associated with the inability to determine which medications or dosage levels were required to reduce reported aggression in residents. Furthermore, data collection also did not include *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition (*DSM*-V) mental health diagnosis beyond HD that might contribute to potentially harmful behaviors.

This is a quality improvement study intended to address the residents' needs in a timely and efficient manner rather than establishing cause and effect between intervention and outcome. The chart audit was used as a proxy for exploring behavior change among the residents, but because they were not direct observations, it is possible that aggressive behaviors were over- or underreported. Furthermore, the first author's selection of words to indicate aggression could be biased. However, these were selected to capture potentially harmful resident behavior. It is possible there are other terms not selected that reflect harmful behavior not included in the analysis.

Conclusion

Employees who care for a population with HD struggle to provide safe, confident, and knowledge-based care. This project demonstrates that employees who participated in a de-escalation training reported improvements in their ability to care for HD residents. There were reductions in residents refusing care and documentation of threatening or aggressive behaviors. Training for employees also resulted in employees feeling safe in their job, increasing knowledge of HD symptoms, and confidence in caring for HD residents themselves and their co-workers. Future research might explore sustainability of de-escalation training in LTC facilities caring for residents with HD.

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Author Contributions

All authors contributed to the conception or design of the study or to the acquisition, analysis, or interpretation of the data. All authors drafted the manuscript, or critically revised the manuscript, and gave final approval of the version that was submitted for publication. All authors agree to be accountable for all aspects of the work, ensuring integrity and accuracy.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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ORCID iDs

Barbara St. Marie https://orcid.org/0000-0003-0231-9464
Daniel Wesemann https://orcid.org/0000-0001-6635-9188

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